SECURELINK VPN LAB

Research Project | Secure VPN for Remote Access

VPN Client Configuration Guide

Establishing a Secure VPN for Remote Access

Deliverable D3 of 6

Organisation	Research project (lab build)
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Version	1.0 (final)
Basis	Lab build on VMware; evidence IDs [E01]-[E38] refer to the evidence screenshot set
Status	Lab documentation. Not certified and not presented as production-ready.

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1 Client matrix

Platform	Software	Status
Linux (Kali 2026.1)	OpenVPN 2.7.5 with client.ovpn and vpn-connect.sh	Implemented and verified [E05][E14]
Windows	OpenVPN GUI	Tested (see D3A)
macOS	Tunnelblick or OpenVPN Connect	Recommended, not tested

2 Linux client: step-by-step

1. Install OpenVPN version 2.6.0 or later (the compliance script enforces this; the lab client runs 2.7.5).
2. Place the profile client.ovpn [E04] and the certificate and key files it refers to in the client directory. Private key permissions must be 600 or stricter [E15].
3. Make sure the system clock is synchronized and a firewall is present; both are checked by vpn-connect.sh [E13][E14].
4. Run vpn-connect.sh. If a check fails, the connection is blocked [E13]; if all checks pass, the VPN starts [E14].
5. When prompted, enter the user name vpnuser and type the current TOTP code from the authenticator app at the password prompt [E05][E09].
6. Verify: the client log shows the server certificate verified, TLS 1.3 with TLS_AES_256_GCM_SHA384 and AES-256-GCM, and one pushed route 10.10.1.0/24 [E05]; the routing table has that single VPN route [E06]; 10.10.1.1 answers ping [E12].

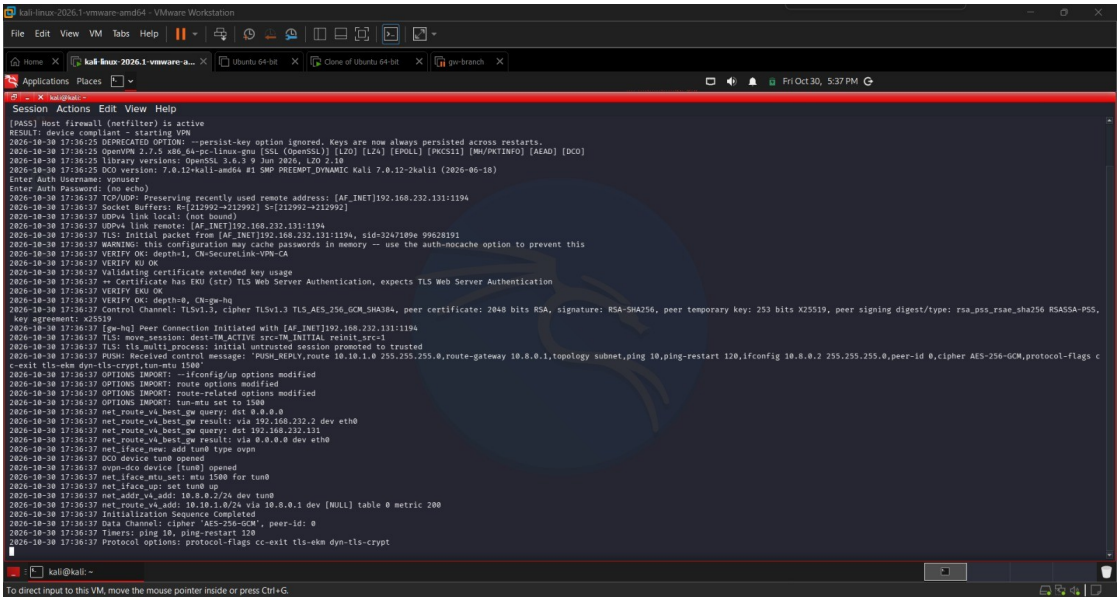


Figure 1. Compliant client connects; TLS 1.3, AES-256-GCM, one pushed route [E05]

Load-balancing profile: client-lb.ovpn connects to either instance (UDP 1194 or 1195) [E21][E22]. The choice is made by the client; there is no automatic failover.

3 Client profile explained

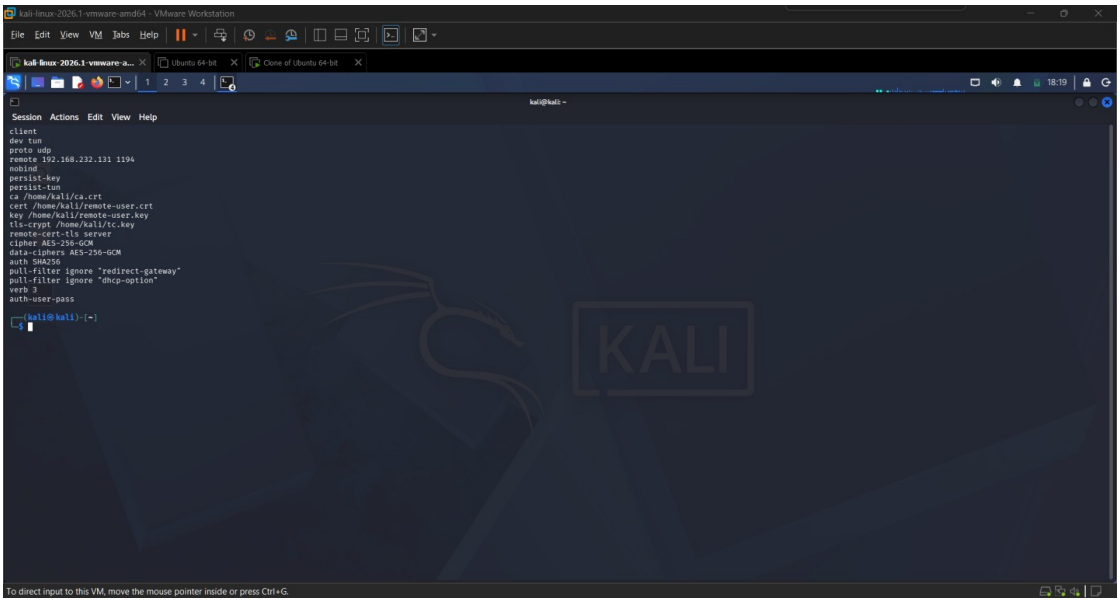


Figure 2. client.ovpn client profile [E04]

Behaviour	Meaning
Server certificate is verified	The client checks the gateway identity before sending credentials [E05]
Ignores redirect-gateway and dhcp-option	Split tunnel: only 10.10.1.0/24 is sent through the VPN; other traffic uses the local network [E06]
Certificate plus password prompt	Client certificate and TOTP code are both required (see D2)
Hardening notes from client logs	persist-key is deprecated; auth-nocache is not set

The complete profile is shown in Figure 2; the behaviours above are the ones that matter for security.

4 Device compliance check

Check (vpn-connect.sh)	Rule
Key file permissions	600 or stricter
OpenVPN version	2.6.0 or later
Clock	System clock synchronized
Firewall	A firewall is present

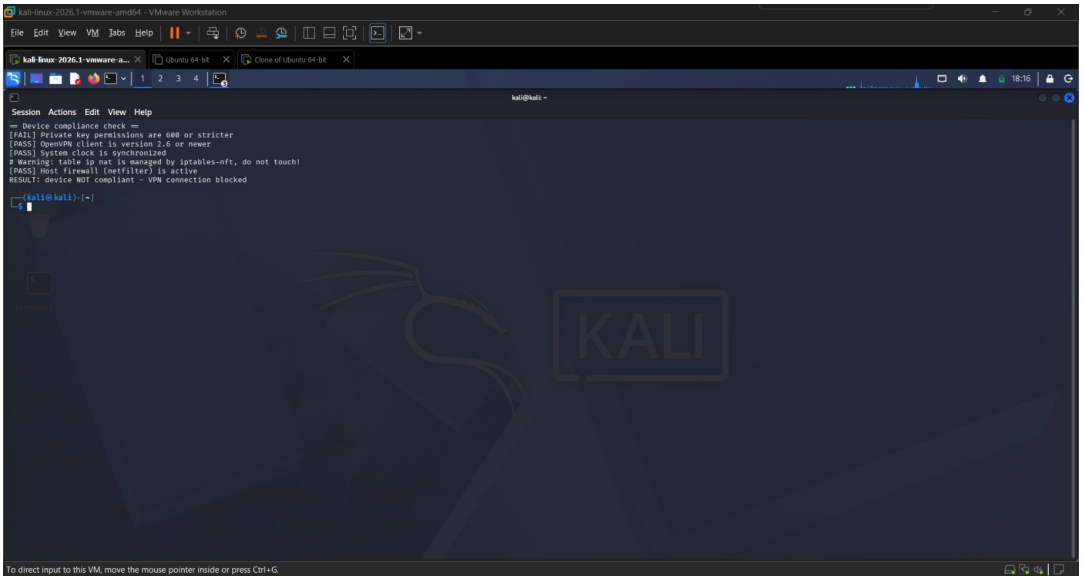


Figure 3. Compliance check fails and the connection is blocked [E13]

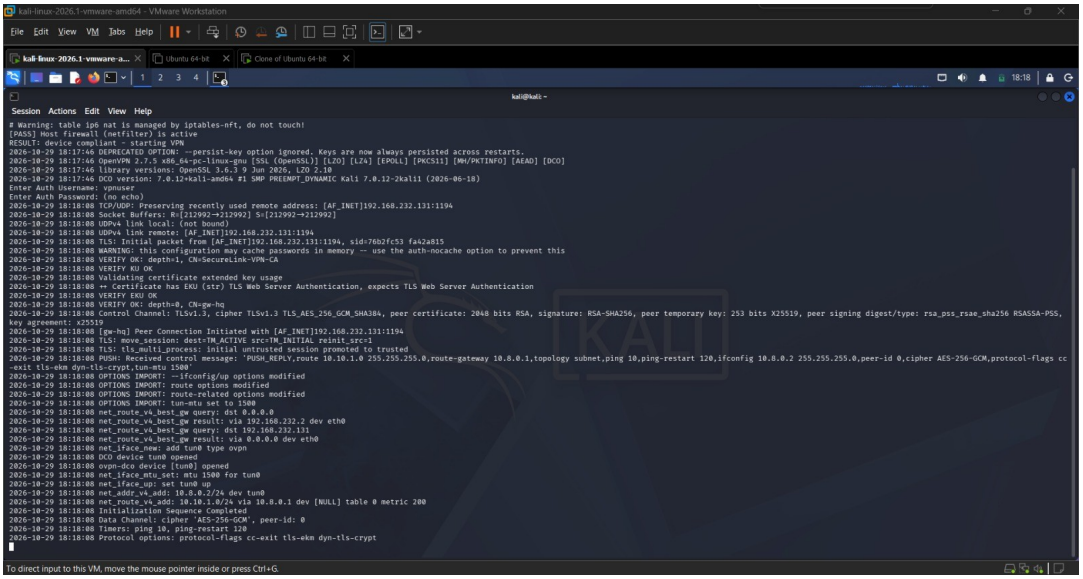


Figure 4. Compliance check passes and the VPN starts [E14]

Limits: the script runs on the client (Linux only) and can be bypassed by starting OpenVPN directly. A server-enforced posture check is recommended, not implemented. A scripted Windows/macOS compliance check was not built (Windows posture was checked manually, see D3A).

5 Troubleshooting from real messages

Message / symptom	Meaning	Action	Evidence
AUTH_FAILED (client); "Invalid verification code" (server)	Wrong or expired TOTP code	Wait for a new code, check the device clock, retry	[E08][E09]
TLS key negotiation failed; "certificate revoked"	Client certificate is on the CRL	Request a new certificate from the administrator	[E24][E25]
Connection blocked before	A compliance check failed	Correct the failed item	[E13]

start		(permissions, version, clock, firewall) and rerun	
Traffic to 10.10.2.1 times out	Destination not allowed by firewall policy	Expected: only 10.10.1.0/24 is permitted	[E11][E12]

Note: the Kali clock is UTC minus 4 hours, so client log times differ from server logs by 4 hours.

6 Windows (tested) and macOS (not tested)

Windows was tested in the lab; see Appendix D3A (D3A_Windows_OpenVPN_Test_Report) and evidence [E29] to [E38]. The steps below are the standard procedure; macOS was not tested and should be checked on a real device before rollout.

7. Install OpenVPN GUI (Windows) or Tunnelblick / OpenVPN Connect (macOS).
8. Import the profile supplied by the administrator, with the certificate and key files it references.
9. Connect and enter the user name and the current TOTP code at the password prompt.
10. Check that only the 10.10.1.0/24 route is added and that 10.10.1.1 is reachable.

7 Implementation status

Category	Items
Implemented and verified	Linux client connection with certificate and TOTP [E05]; split tunnel [E06]; compliance fail and pass [E13][E14]
Implemented but limited	Compliance check is client-side, Linux-only and bypassable; load-balancing profile is client-chosen
Recommended, not implemented	Tested macOS client; server-enforced posture checks; remove persist-key and set auth-nocache